

Task 22 – Cohort Retention Analysis

Tools: Python, Pandas, Matplotlib, Seaborn, Google Colab

Dataset: Online Retail II

1. Objective

The objective was to perform customer cohort retention analysis by grouping customers according to the month of their first purchase, tracking activity in subsequent months, calculating retention percentages, and visualizing the results using a cohort heatmap.

2. Data Preparation

The Online Retail II Excel dataset was loaded from the 2009–2010 and 2010–2011 sheets and combined. Invoice dates were converted to datetime. Records without Customer ID, cancelled invoices, non-positive quantities, and duplicate records were removed.

3. Cohort Methodology

For each customer, the earliest purchase date was identified. Its month became the CohortMonth. Each transaction was assigned an ActivityMonth. CohortIndex represents the number of months since first purchase, with the original month as Month 1.

4. Retention Calculation

Unique active customers were counted for each CohortMonth and CohortIndex. Retention percentage was calculated as active customers divided by the cohort's Month 1 customers, multiplied by 100.

5. Key Results

Metric	Result
Month 1 retention	100%
Average Month 2 retention	21.16%
Average Month 3 retention	21.92%
Average Month 4 retention	21.61%
Average Month 11 retention	14.99%
Largest cohort	December 2009 – 955 customers

6. Business Insights

- Month 1 is 100% because it represents the original cohort purchase month.
- Average retention drops substantially after the first month, with Month 2 at 21.16%.
- Retention remains around 21% through Months 2–4 and declines further over longer periods.
- The December 2009 cohort was the largest, with 955 customers.
- Later cohorts have fewer observation months and should not be directly compared with older cohorts for long-term retention.

7. Visualization

A customer cohort retention heatmap was created in Google Colab and saved as **cohort_retention_heatmap.png**.

8. Conclusion

The analysis shows a clear drop in customer retention after the initial purchase month, followed by lower engagement in later months. The cohort heatmap helps identify retention patterns across customer acquisition cohorts.